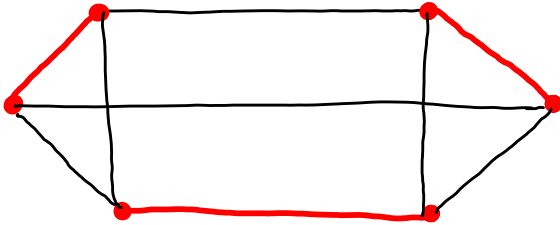


Chapter 5

Factors and Factorisations

5.1 Tutte's 1-Factor Theorem

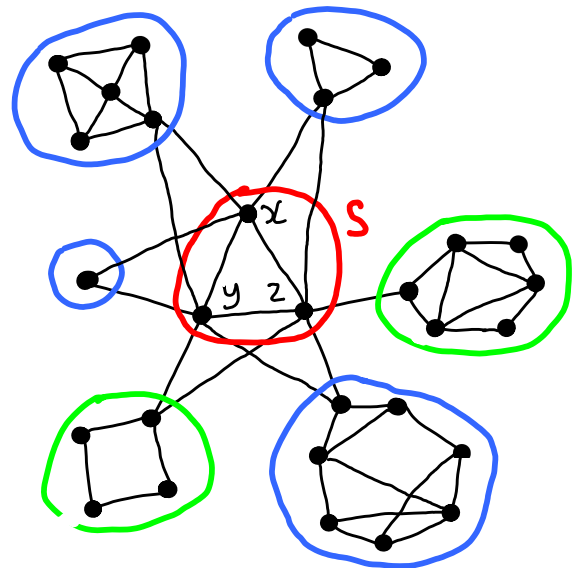
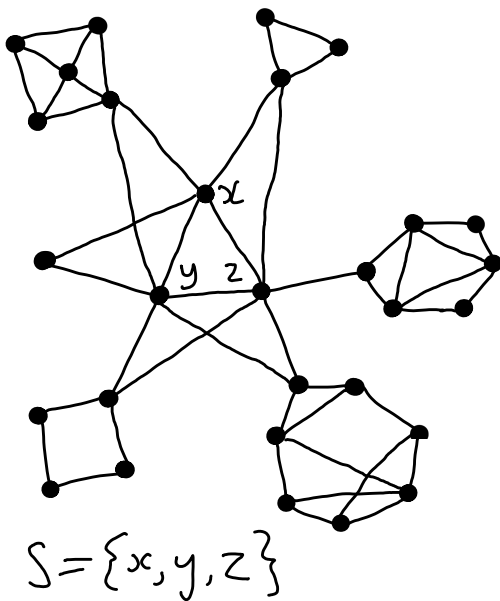
A spanning subgraph is called a **factor**, a k -regular factor is called a **k -factor**



1-factor
(perfect matching)

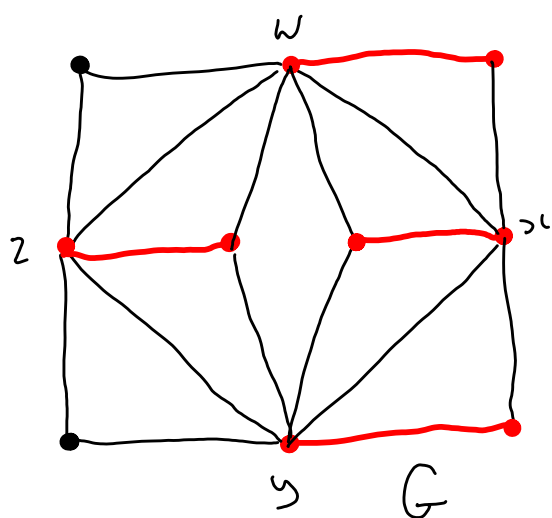
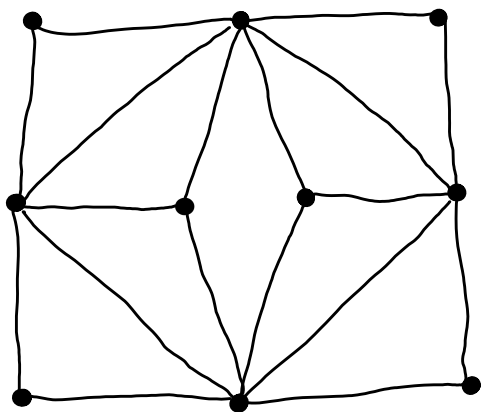
The following theorem was proved by Tutte in 1947 [18].

Theorem 5.1.1. (Tutte's 1-Factor Theorem) A graph G has a 1-factor if and only if for every $S \subseteq V(G)$ the number of components of odd order in $G - S$ does not exceed $|S|$.



$G - S$ has 4 components
of odd order, but $|S| = 3$.

Exercise: Find a largest 1-regular subgraph of the graph below.

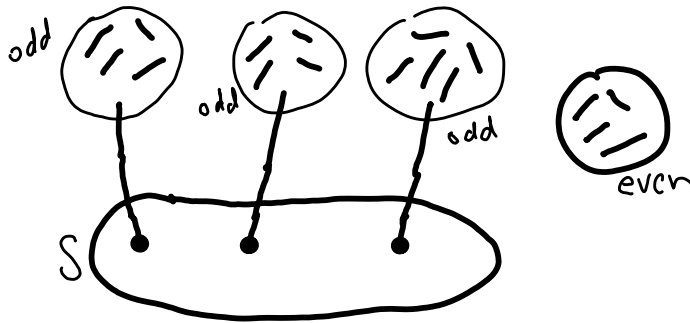


The graph does not have a 1-factor because $G - S$, where $S = \{w, x, y, z\}$, has 6 components of odd order but $|S| = 4$. Thus, there is no 1-regular subgraph that is larger than the one shown in red.

Theorem 5.1.1. (Tutte's 1-Factor Theorem) A graph G has a 1-factor if and only if for every $S \subseteq V(G)$ the number of components of odd order in $G - S$ does not exceed $|S|$.

Proof Let $o(G)$ denote the number of components of odd order in a graph G . So we need to show that a graph G has a 1-factor if and only if $o(G - S) \leq |S|$ for every $S \subseteq V(G)$. We only need to prove the theorem for simple graphs because multiple edges make no difference to the existence of a 1-factor, nor to the condition that $o(G - S) \leq |S|$ for every $S \subseteq V(G)$. Tutte condition

If G has a 1-factor F , then for any component H of odd order in $G - S$, there is an edge of F joining a vertex of H to a vertex in S . It follows that $o(G - S) \leq |S|$.

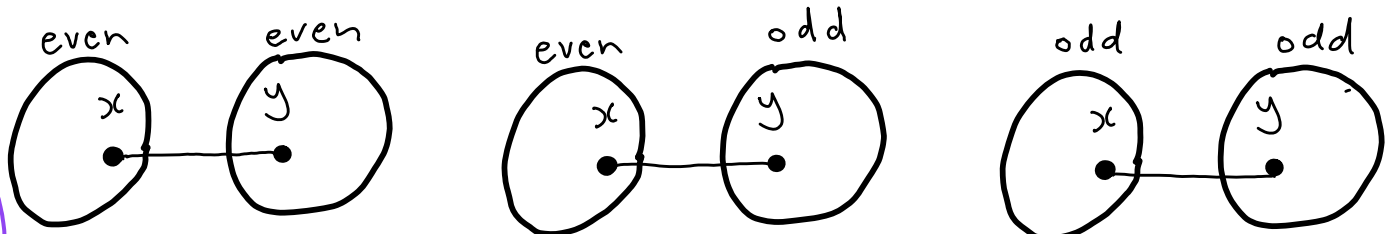


We now proceed to show that if $o(G - S) \leq |S|$ for every $S \subseteq V(G)$, then G has a 1-factor. For a contradiction, suppose that there exists a counterexample G_0 to this statement. That is, G_0 is a graph such that $o(G_0 - S) \leq |S|$ for every $S \subseteq V(G_0)$, but G_0 has no 1-factor. Taking $S = \emptyset$, we see that G_0 has even order. Clearly, complete graphs of even order have 1-factors so G_0 is not complete.

If there exist non-adjacent vertices x and y in G_0 such that $G_0 + xy$ has no 1-factor, then $G_0 + xy$ is another counterexample. This is because joining non-adjacent vertices of a graph does not increase the number of odd components. So $G_1 = G_0 + xy$ satisfies $o(G_1 - S) \leq |S|$ for every $S \subseteq V(G_1)$ and G_1 has no 1-factor. That is, G_1 is also a counterexample. Thus, there exists a counter-example G such that if x and y are any non-adjacent vertices in G , then $G + xy$ has a 1-factor.

If $x \in S$ or $y \in S$, then $G_0 + xy - S = G_0 - S$.

If x, y are in the same component of $G_0 - S$, then $o(G_0 - S) = o(G_0 + xy - S)$.



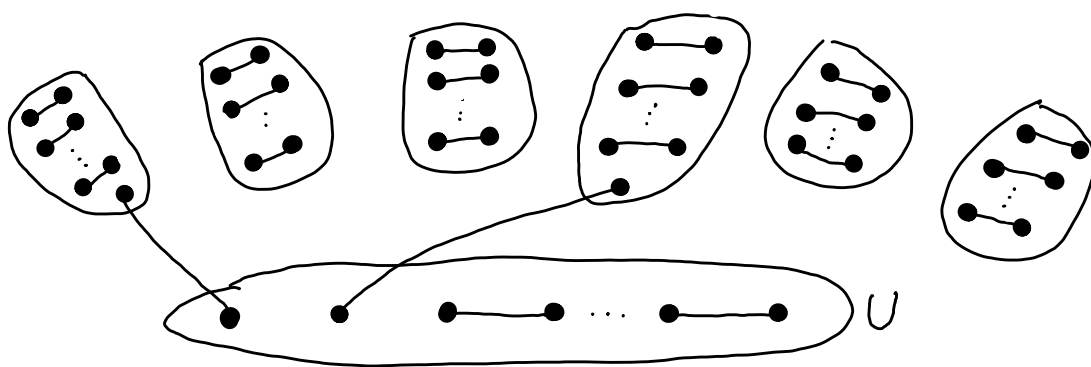
$$o(G_1 - S) = o(G_0 + xy - S) \leq o(G_0 - S) \leq |S|$$

explanation that adding edge xy does not increase the number of odd components.

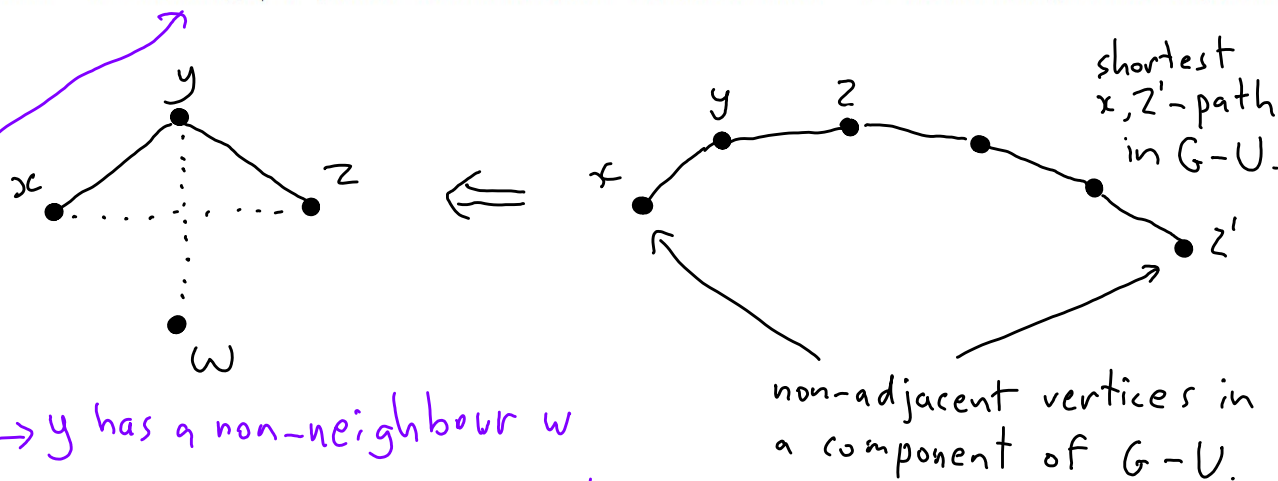
Let U be the set of vertices of G that have degree $n(G) - 1$ (so a vertex in U is adjacent to every vertex of G , except itself). The proof now splits into two cases: the case where every component of $G - U$ is complete, and the case where there is a non-complete component in $G - U$.

① Suppose first that every component of $G - U$ is complete. Any complete graph of even order has a 1-factor, and any complete graph of odd order has a 1-regular subgraph that contains every vertex except one. Thus, there is a 1-regular subgraph F_1 of $G - U$ that contains all the vertices of $G - U$ except for one vertex from each odd-order component of $G - U$.

Since $o(G - U) \leq |U|$ and since every vertex in U is adjacent to every vertex of G (except itself), there is a 1-regular subgraph F_2 of G containing the $o(G - U)$ vertices of $G - U$ that are not in F_1 , and containing $o(G - U)$ vertices from U . Since $n(G)$ and $n(F_1 \cup F_2)$ are both even, $|V(G) \setminus V(F_1 \cup F_2)|$ is even, and by the definition of U the vertices in $V(G) \setminus V(F_1 \cup F_2)$ are pairwise adjacent. Thus, $F_1 \cup F_2$ can be extended to a 1-factor of G .



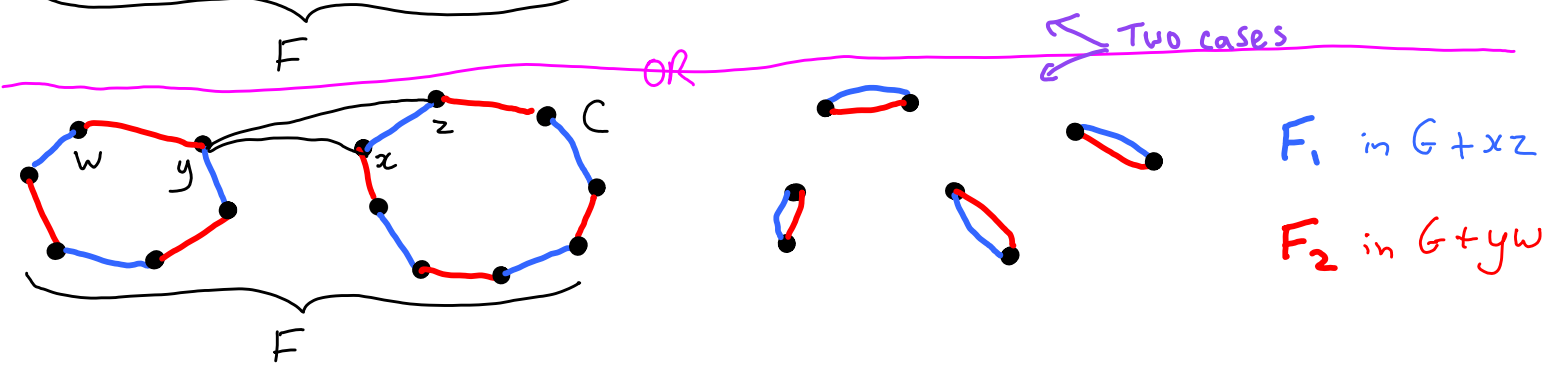
② Now suppose that there is a non-complete component in $G - U$. Thus, $G - U$ contains vertices x and z that are at distance 2 (any non-complete connected graph contains non-adjacent vertices, which are at distance at least 2, and hence contains non-adjacent vertices at distance exactly 2). Let $[x, y, z]$ be a path in $G - U$. Since $y \notin U$, there exists another vertex w in $G - U$ such that y is not adjacent to w .



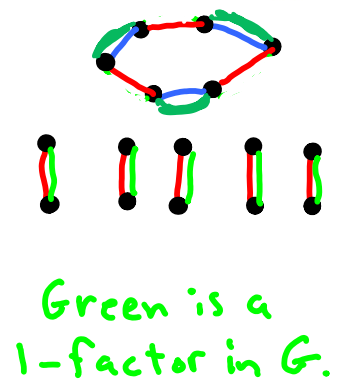
$y \notin U \rightarrow y$ has a non-neighbour w

w has a non-neighbour $(y) \rightarrow w \notin U$ by definition of U .

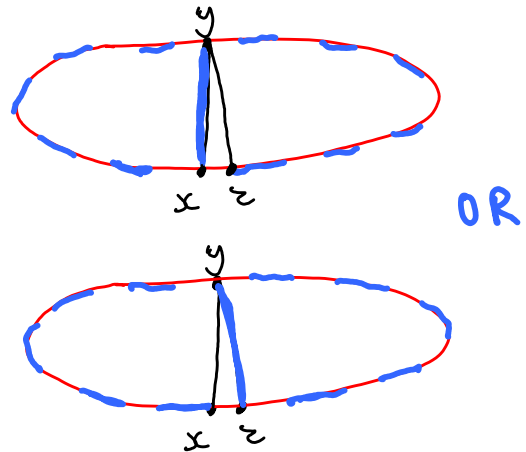
By the definition of G , the graphs $G + xz$ and $G + yw$ each have 1-factors. Let F_1 be a 1-factor in $G + xz$ and let F_2 be a 1-factor in $G + yw$. Thus, xy and yz are not in F_1 , and xy and yz are not in F_2 . Let F be the subgraph of G induced by the edges in exactly one F_1 and F_2 . Note that xz and yw are edges of F and that each component of F is a cycle of even length. Let C be the cycle of F that contains the edge xz .



If C does not contain yw , then the required 1-factor of G consists of the edges in F_2 that are in C and all the edges of F_1 that are not in C . Thus, we can assume that xz and yw are both in C . It is easy to see that $C \cup [x, y, z]$ has a 1-factor that contains neither of the edges xz nor yw . The edges of this 1-factor together with the edges of F_1 that are not in C form a 1-factor in G . This is a contradiction and the proof is complete. $\square$



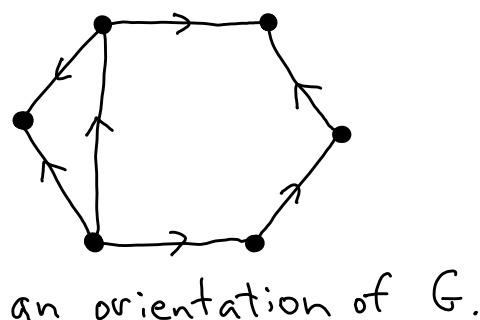
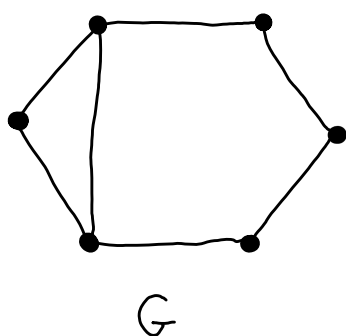
Exercise: Draw some other possibilities for the case where xz and yw are both in C , and convince yourself that in each case $C \cup \{x, y, z\}$ has a 1-factor that contains neither xz nor yw .



(one of xy and zy
"divides" the cycle
into two "even" cycles)

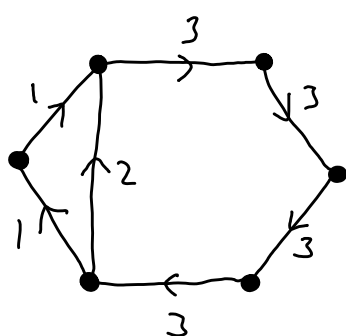
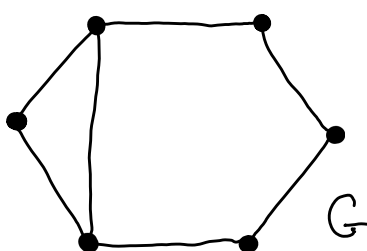
5.2 Flows in Graphs

Definition 5.2.1. An **orientation** of a graph G is a digraph obtained from G by assigning an orientation, either x to y or y to x , to each edge xy of G .

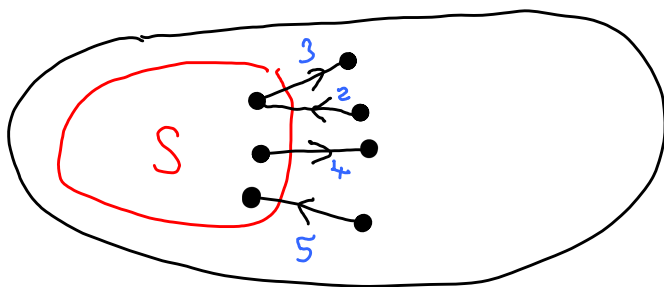


Definition 5.2.2. A **flow** on a graph G is a pair (D, f) where D is an orientation of G and f is an assignment of weights to the edges of D such that for each vertex $v \in V(G)$, the sum of the weights on the edges directed into v equals the sum of the weights on the edges directed out of v .

Exercise: Find a flow on the graph G .



Lemma 5.2.3. For any flow (D, f) on a graph G , the net flow out of any subset $S \subseteq V(G)$ is zero, where the net flow out of S means the sum of the flows on the edges leaving S minus the sum of the flows on the edges entering S .

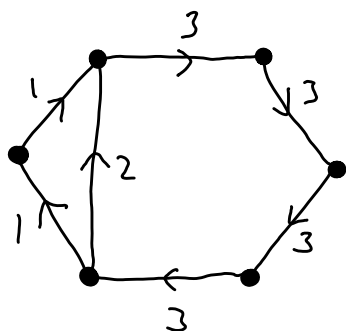
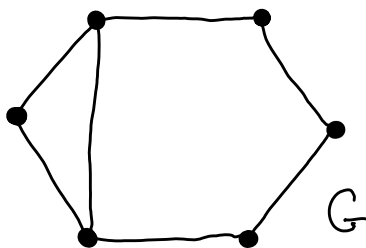


Proof By Lemma 3.1.3 the net flow out of S is the sum of the net flows out of the vertices of S , and the net flow out of each vertex is zero by definition. $\square$

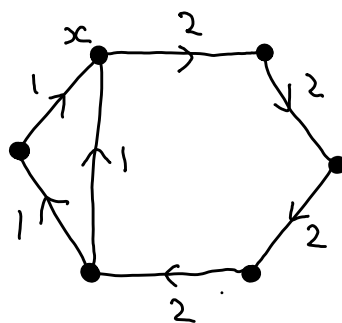
Definition 5.2.4. Let $k \geq 2$ be an integer. A **nowhere-zero flow** is a flow in which no weight is zero, and a **nowhere-zero k -flow** is a flow in which all of the weights are in $\{1, 2, \dots, k-1\}$.

Exercise: What is the minimum k such that G has a nowhere-zero k -flow?

Find a nowhere-zero k -flow for the minimum k .



A nowhere-zero 4-flow on G .

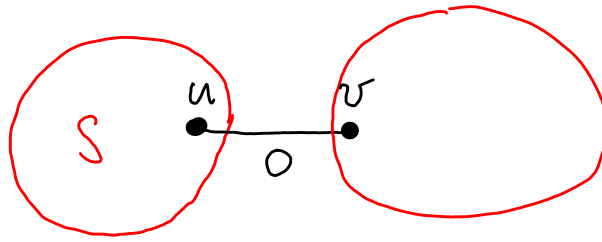


A nowhere-zero 3-flow on G .

Since x has degree 3, it is not possible that all the edges incident on x have weight 1. Thus, $k=3$ is a minimum.

Theorem 5.2.5. If a graph has a nowhere-zero flow, then it has no bridges.

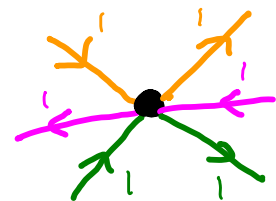
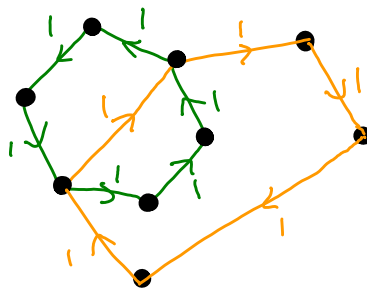
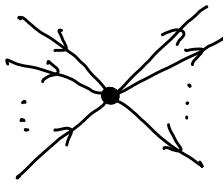
Proof Let $e = uv$ be a bridge in a graph G and suppose there is a flow on G . By Lemma 5.2.3, there is zero net flow out of the vertex set of the component of $G - \{e\}$ containing u . Thus, the flow on e is zero. $\square$



A famous unsolved conjecture of Tutte [19] states that every bridgeless graph has a nowhere-zero 5-flow. Seymour [15] has shown that every bridgeless graph has a nowhere-zero 6-flow.

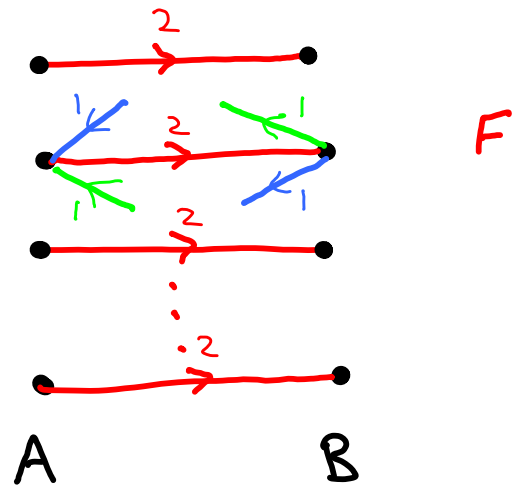
Theorem 5.2.6. A graph has a nowhere-zero 2-flow if and only if each vertex has even degree.

Proof In a nowhere-zero 2-flow each edge has weight 1 and it follows that the indegree of each vertex equals its outdegree. Thus, each vertex has even degree. Conversely, if each vertex of a graph has even degree, then by Theorem 1.0.2 there is a decomposition of G into cycles. Clearly, any cycle has a nowhere-zero 2-flow. If we assign weights and orientations so that each cycle in the decomposition has a nowhere-zero 2-flow, then we obtain a nowhere-zero 2-flow in the graph. $\square$

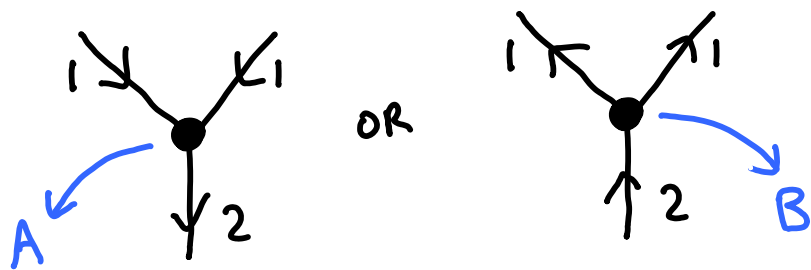


Theorem 5.2.7. A 3-regular graph has a nowhere-zero 3-flow if and only if it is bipartite.

Proof Let G be a bipartite 3-regular graph and let A and B be the parts. By Theorem 1.0.3, G has a proper 3-edge colouring. Let F be the 1-factor corresponding to one of the colour classes. Each edge of F is oriented from A to B and given weight 2. The other edges are oriented from B to A and given weight 1. This is easily seen to be a nowhere-zero 3-flow.



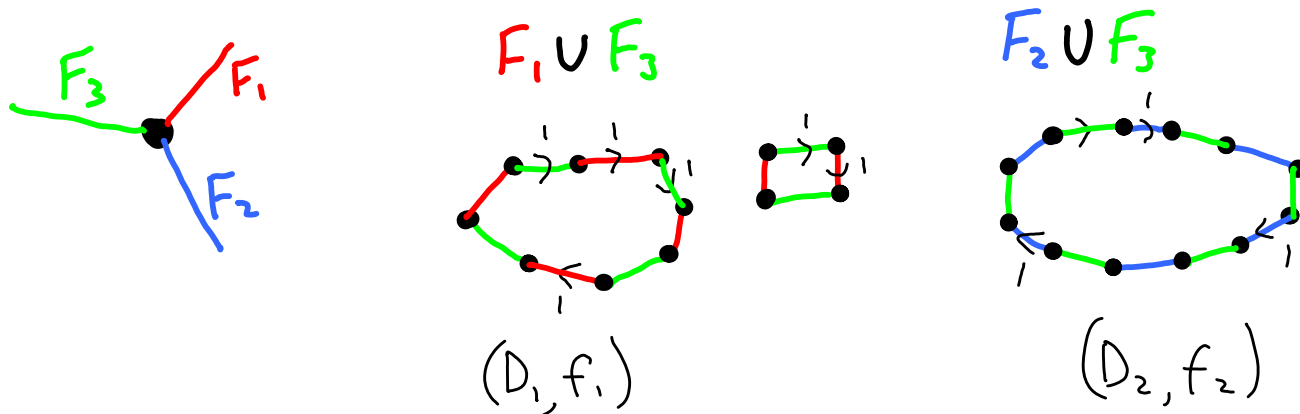
Conversely, let G be a 3-regular graph with a nowhere-zero 3-flow. It is clear that each vertex is incident with an edge of weight 2 and two edges of weight 1. If we let A be the set of vertices incident with the tail of a weight 2 edge, and let B be the set of vertices incident with the head of a weight 2 edge, then each edge of weight 1 has its tail in B and its head in A . Thus, G is bipartite. $\square$



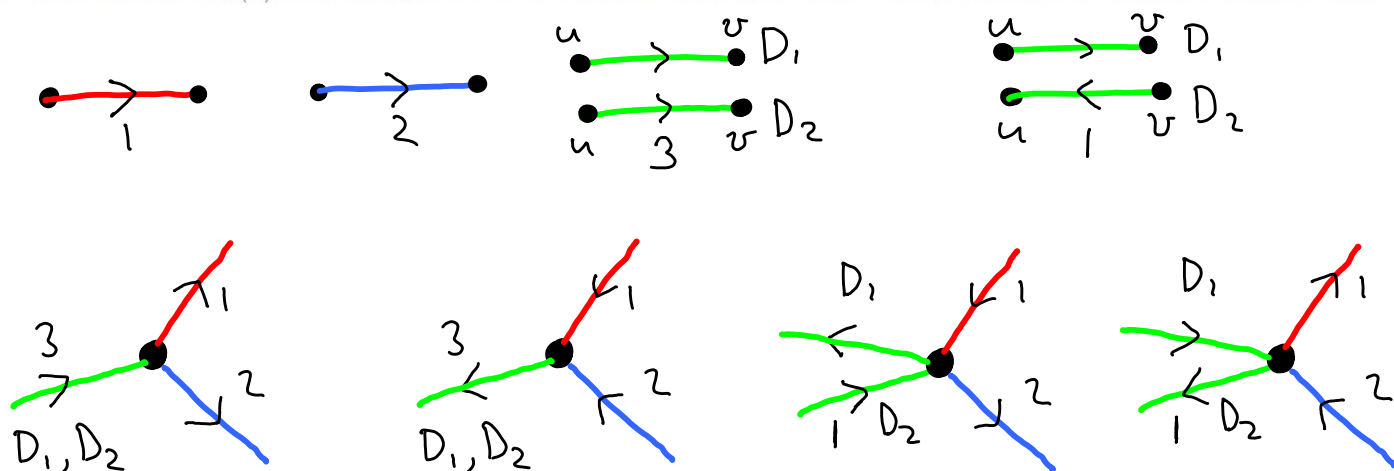
Theorem 5.2.8. A 3-regular graph has a nowhere-zero 4-flow if and only if it has a 1-factorisation.

equivalent to a proper 3-edge colouring ↗

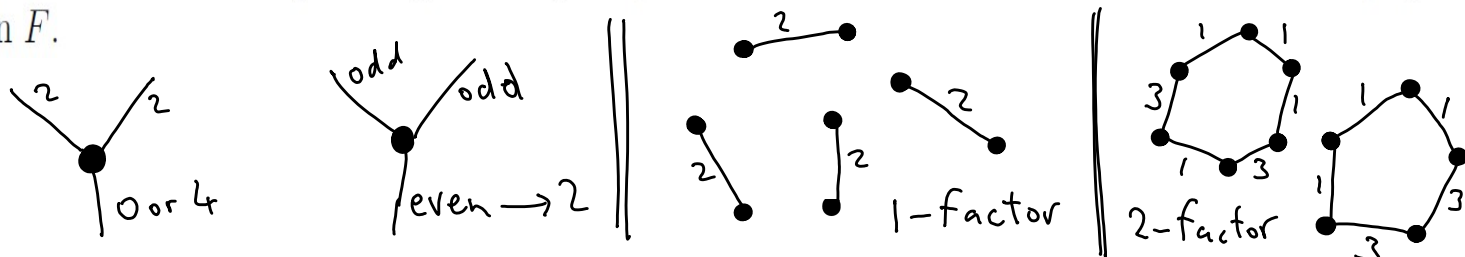
Proof Let G be a 3-regular graph with a 1-factorisation $\{F_1, F_2, F_3\}$. Then each of $F_1 \cup F_3$ and $F_2 \cup F_3$ is a 2-factor. Any 2-factor has a nowhere-zero 2-flow. Let (D_1, f_1) be a 2-flow on $F_1 \cup F_3$ and let (D_2, f_2) be a 2-flow on $F_2 \cup F_3$.



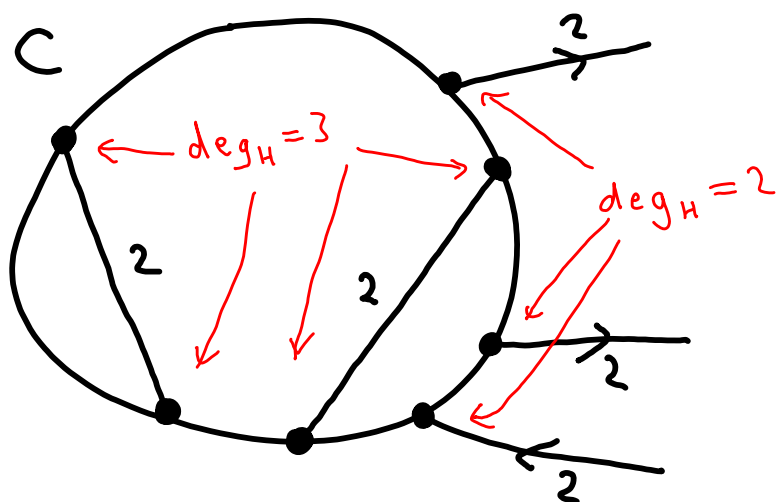
Define a flow on G as follows. If e is an edge of F_1 , then e has orientation $D_1(e)$ and weight 1. If e is an edge of F_2 , then e has the orientation $D_2(e)$ and weight 2. If e is an edge of F_3 and $D_1(e) = D_2(e)$, then e has the orientation $D_1(e)$ and weight 3. If e is an edge of F_3 and $D_1(e) \neq D_2(e)$, then e has the orientation $D_2(e)$ and weight 1. It is routine to check that this produces a nowhere-zero 4-flow.



Conversely, let G be a 3-regular graph with a nowhere-zero 4-flow. It is easy to see (using parity arguments) that each vertex is incident with exactly one edge of weight 2. Thus, the edges of weight 2 form a 1-factor of G . The edges of weight 1 or 3 thus form a 2-factor F of G . It suffices to show that every cycle of F has even length, because then F has a 1-factorisation and this together with the 1-factor formed by the edges of weight 2 yields a 1-factorisation of G . Let C be an arbitrary cycle in F .



Consider the subgraph H of G induced by the vertices of C . A vertex of H has degree 3 in H if the incident edge of weight 2 is in H , and has degree 2 in H if the incident edge of weight 2 is not in H . Thus, H has an even number of vertices of degree 3, and so it suffices to show that there are an even number of vertices of degree 2 in H . Now consider the net flow out of $V(C)$, which is zero



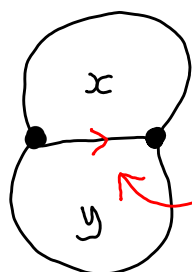
Want to show
 C has even
length.

by Lemma 5.2.3. The net flow out of $V(C)$ is also equal to twice the number of edges of weight 2 leaving $V(C)$ minus twice the number of edges of weight 2 entering $V(C)$. These two numbers are thus equal and so there are an even number of vertices of degree 2 in H . Hence C has even length and G has a 1-factorisation. $\square$

Theorem 5.2.9. Let G be a plane bridgeless graph. Then G is k -face-colourable if and only if it has a nowhere-zero k -flow.

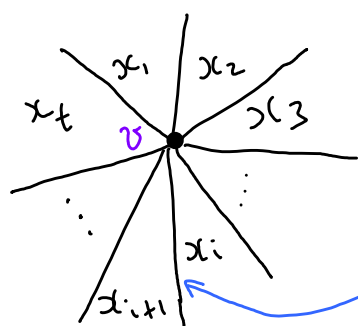
Proof: ① k -face colourable $\longrightarrow$ nowhere-zero k -flow.

Can assume colours are $0, 1, \dots, k-1$ and external face is 0 .



orient so that the larger of x and y is on the left and give weight "left colour - right colour".
So $1 \leq x - y \leq k - 1$.

If it is a flow, then it is a nowhere-zero k -flow.



contributes $x_{i+1} - x_i$ to the flow into v .

① $x_{i+1} > x_i$

$\uparrow x_{i+1} - x_i$

② $x_i > x_{i+1}$

$\downarrow -(x_i - x_{i+1}) = x_{i+1} - x_i$

Net flow into v

$$= (x_2 - x_1) + (x_3 - x_2) + (x_4 - x_3) + \dots + (x_t - x_{t-1}) + (x_1 - x_t) \\ = 0$$

So it is a flow.

② nowhere-zero k -flow $\rightarrow k$ -face colourable.

Assign colour 0 to external face.

edge left-to-right $\Rightarrow +$ weight
edge right-to-left $\Rightarrow -$ weight.

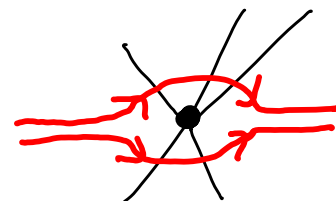
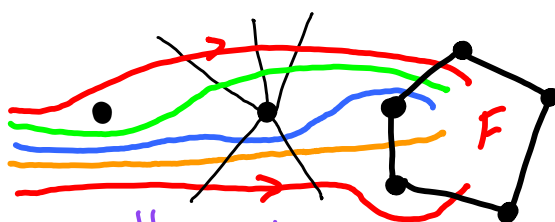
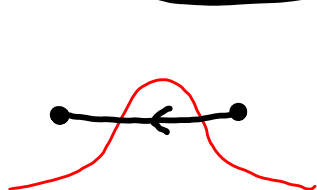
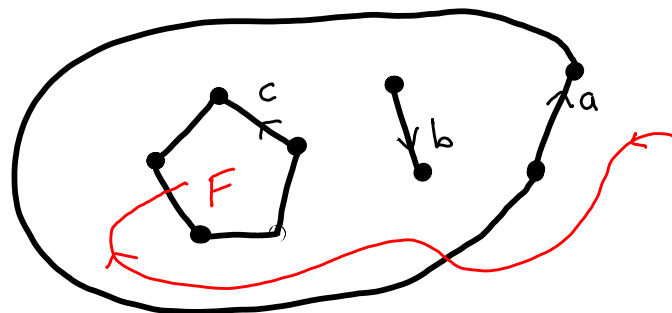
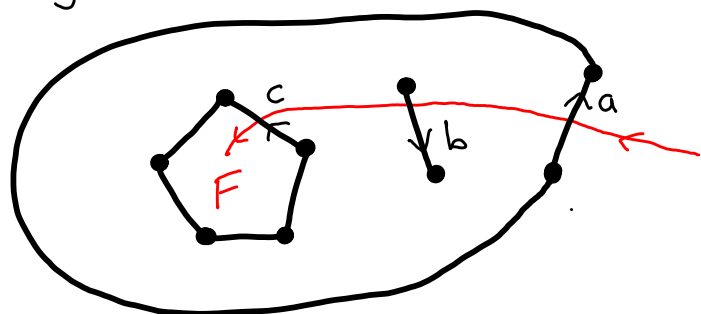
Then reduce modulo k on $\{0, 1, \dots, k-1\}$

$$g(F) = a - b + c.$$

adjacent faces differ by $1, 2, \dots, k-2, k-1$.
(So distinct mod k)

We need to show that g is well defined.

If g is well defined, then it is a proper k -face colouring.



any path can be converted to any other path by going the other way around vertices.

$$\Sigma_1 = -a + b - c$$

$\Sigma_1 =$ flow into v

Net flow into $v = \Sigma_1 - \Sigma_2$.

$$\text{So } \Sigma_1 = \Sigma_2$$

$\Sigma_2 =$ flow out of v

$$\Sigma_2 = -e + d$$

So g is well-defined.



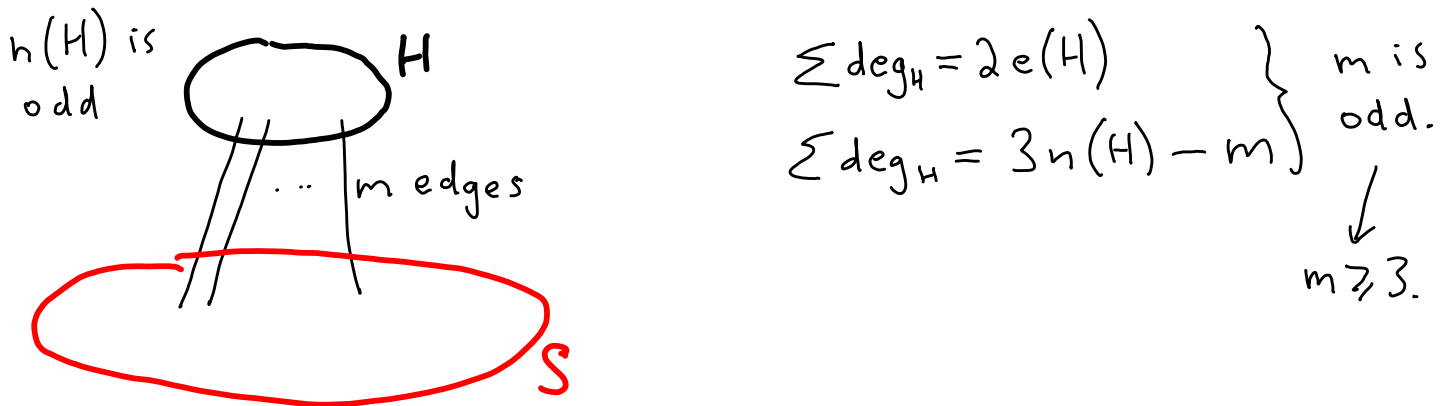
5.3 Two Theorems of Petersen

The following two theorems were proved by Petersen in 1891 [13]. The first, Theorem 5.3.1, is generally known as “Petersen’s Theorem”. Our proof of Theorem 5.3.1 uses Tutte’s 1-Factor Theorem, which was not proved until 1947.

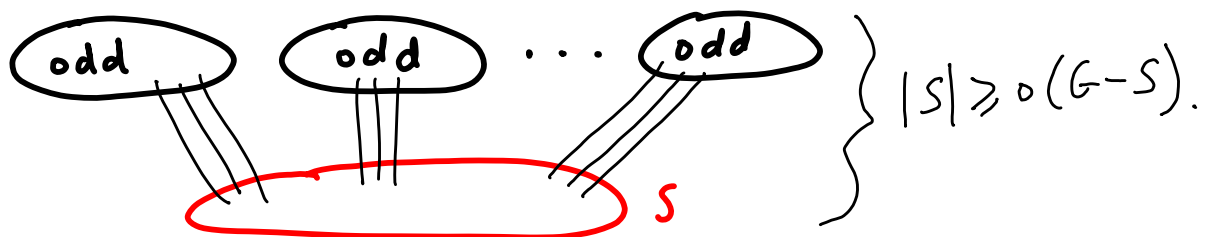
Theorem 5.3.1. Every bridgeless 3-regular graph has a 1-factor

Proof We use Tutte’s 1-Factor Theorem (Theorem 5.1.1) to prove the result. Let G be a bridgeless 3-regular graph, let $S \subseteq V(G)$, let $o(G - S)$ denote the number of components of odd order in $G - S$, let H be a component of odd order in $G - S$, and let m be the number of edges joining H to S . The sum of the degrees of the vertices of H is $2e(H)$ and is also $3n(H) - m$. Thus, since $n(H)$ is odd, m is also odd. Since G has no bridges, this means that $m \geq 3$.

Theorem 5.1.1. (Tutte’s 1-Factor Theorem) A graph G has a 1-factor if and only if for every $S \subseteq V(G)$ the number of components of odd order in $G - S$ does not exceed $|S|$.

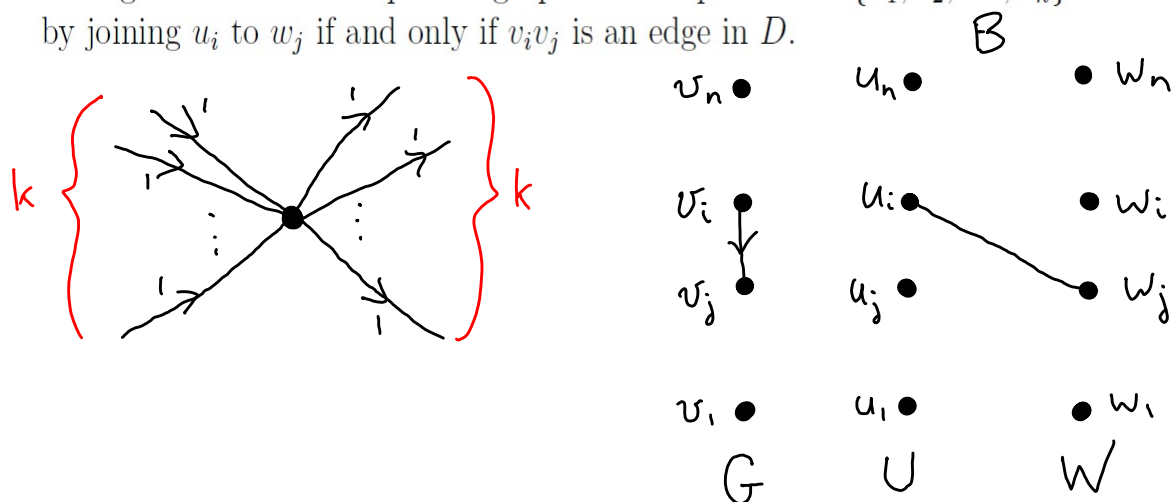


Thus, the number of edges of G that join odd components of $G - S$ to S is at least $3o(G - S)$. Since each vertex of S is incident with at most 3 such edges, this implies that $|S| \geq o(G - S)$. Thus, by Tutte’s 1-Factor Theorem, G has a 1-factor. $\square$

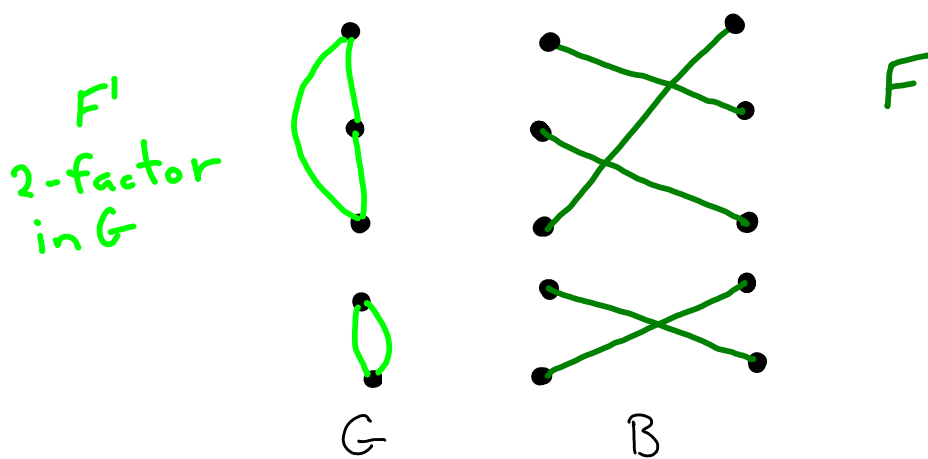


Theorem 5.3.2. Every regular graph of positive even degree has a 2-factor, and every regular graph of even degree has a 2-factorisation.

Proof Let G be a regular graph of positive even degree $2k$ and let the vertices of G be $v_1, v_2, \dots, v_n$. By Theorem 5.2.6, G has a nowhere-zero 2-flow, and thus there is an orientation D of G in which the indegree of each vertex is equal to its outdegree. That is, each vertex of D has indegree k and outdegree k . Define a bipartite graph B with parts $U = \{u_1, u_2, \dots, u_n\}$ and $W = \{w_1, w_2, \dots, w_n\}$ by joining u_i to w_j if and only if $v_i v_j$ is an edge in D .



Since the outdegree of each vertex of D is k , the degree in B of each vertex of U is k . Since the indegree of each vertex of D is k , the degree in B of each vertex of W is k . That is, B is k -regular. Thus, by Theorem 1.0.3, B is k -edge colourable. Let F be a 1-factor of B induced by one of the colour classes in a proper k -edge colouring of B . If we define a subgraph F' of G by letting $v_i v_j$ be an edge in F' if and only if $u_i w_j$ is an edge in F , then F' is a 2-factor of G .



The second part of the theorem, that every regular graph of even degree has a 2-factorisation, follows immediately from the first part of the theorem because removing the edges of a 2-factor from a regular graph of positive even degree results in a regular graph of even degree. So the process can be repeated until a 2-factorisation is obtained. Alternatively, the colour classes of the proper k -edge colouring of B define the 2-factors of the required 2-factorisation. $\square$

